Xiaotong (Kary) Fang

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Summary

PhD researcher in Computational Neuroscience at USC. I study how humans learn and make decisions under uncertainty. My work involves developing interpretable machine learning and statistical models—including Hidden Markov Models, particle filters, and compact RNNs—that disentangle different sources of noise (volatility vs. stochasticity) in decision tasks. Using large behavioral datasets ($\tilde{4}$,600 participants) and psychiatric surveys, I test how these uncertainty signatures map onto transdiagnostic symptom dimensions. I also extend these paradigms into immersive VR with pupillometry to link behavior with physiological markers. Beyond neuroscience, I build end-to-end ML pipelines in Python/PyTorch and explore LLM fine-tuning for simulating psychological traits, with the broader aim of creating scalable and interpretable AI systems that has real-world application.
Available for summer/Fall 2026 internship.

Research & Technical Projects

Modeling Human Learning under Uncertainty Aug 2023 – Oct 2025

- Engineered and deployed **web-based behavioral tasks** using **JavaScript + HTML**, collecting trial-level data from **4,600+ participants** under manipulations of volatility and stochasticity.
- Developed binary-native **inference model** and benchmarked with **Bayesian models** linked uncertainty processing to clinical transdiagnostic traits.
- Applied **factor analysis** on survey data, uncovering a **double dissociation**: internalizing symptoms ↔ stochasticity misestimation, externalizing ↔ volatility misestimation.
- Built **end-to-end ML pipelines** in Python and MATLAB (data cleaning → model fitting → parameter recovery → simulation → visualization) with **GitHub version control** for reproducibility.
- Published **two manuscripts** (one under review, one in revision) establishing a **normative theory of learning under uncertainty** with implications for psychiatric modeling.

Interpretable Tiny RNNs for Decision-Time Modeling July 2025 – Present

- Integrated **reaction time likelihoods** into tiny-RNN training objective, improving **identifiability** vs. drfit diffusion models and bridge gaps in classical decision modeling frameworks.
- Producing **phase portrait and attractor dynamics** for interpretable latent computations.
- Built modular pipelines in **PyTorch with GPU acceleration**, allowing scaling to multimodal datasets (behavioral + physiological) with reproducibility controls.
- Advances **interpretable deep learning** and **data-efficient sequence modeling** for **time-series AI** (forecasting, recommendation, adaptive systems).
- Provides **minimal mechanistic models** for human / animal studies, enabling **computational comparisons between species**.

Immersive VR Task with Pupillometry Sep 2025 – Present

- Designed a **VR-based** immersive task in **Unity** with integrated Tobii eye-tracking, motion tracking, and physiology recording to probe phasic vs. tonic responses to uncertainty.
- Building a **sensor fusion pipeline**: Unity task engine → VR headset interface → Tobii SDK → synchronized multimodal data streams (behavioral, pupillometric, physiological) processed in **Python**.
- Demonstrated **real-time multimodal signal integration** and streaming ML preprocessing, advancing scalable approaches for **AR/VR-based adaptive user modeling**.
- Foundations for **digital biomarkers** of psychiatric risk using scalable, immersive VR tasks.

AI Clinical Assistant for Therapist Support Sep 2025 – Present

- Designing an **LLM-driven interactive agent** that conducts psychological intake interviews, administers tasks, and generates clinician-ready summaries with predicted symptom severity.
- Implementing **conversation state tracking**, IRT scoring, and guardrailed dialogue policies for safe, evidence-grounded patient interactions.
- Developing a **clinician-facing dashboard** with calibrated risk scores, key patient quotes, and suggested follow-up probes to streamline diagnosis.
- Building a roadmap toward **real-time therapist support**, integrating **ASR + diarization** with live guidance and safety-critical alerts.

Topological Masking for Universal Decoding of Spiking Neural Dynamics Aug-Dec 2022

- Highlighted implications for **brain-machine interfaces and neural prostheses**, showing transfer learning across individuals as a path for adaptive neurotech.
- Developed a **universal neural decoder** for hippocampal spike data across 59 rats, aligning subjects via **neural masking** and **attention**, while preserving shared dynamics.
- Improved **cross-subject decoding accuracy to 0.76** vs. 0.65 baseline (**15% lift**); ablations confirmed **attention (+25%)** and masks (+4%) as critical.
- Demonstrated **transfer to unseen subjects**: zero-shot above chance (0.30 vs. 0.25), embedding fine-tuning (0.57), and full fine-tuning (0.79).

Clearance Prediction Modeling Mar 2025

- Built an **end-to-end ML pipeline** (EDA, imputation, outlier handling, scaling, feature engineering) on **520k training records** of booking and pricing data with strong seasonality patterns.
- Trained and compared GBMs under multiple **time-series CV strategies** using MAE and RMSE.
- Achieved best performance with expanding-window CV: 15%+ reduction in error compared to baseline.
- Applied feature selection and SHAP-based interpretability to drop 17 low-importance features, improving generalization and reducing model complexity.
- Proposed **quantile regression extension** to generate confidence intervals, providing more robust predictions for downstream decision-making.

Work Experience

Beijing Topjoy Technology Co. Ltd. – Data Science Intern Beijing, China | May–Sep 2021

- Built a **churn prediction system** on **10M+ user behavior logs**, engineering **temporal engagement features and embeddings** with Python + SQL.
- Benchmarked **XGBoost, LightGBM, and sequence models (RNNs on user sessions)**, achieving **AUC 0.84 vs. 0.71 baseline** (≈18% relative lift) and improving early churn detection.
- **Deployed the best model** as a **batch inference service via REST API on Alibaba Cloud**, enabling product teams to run **weekly churn forecasts at scale**.
- Automated **reporting dashboards** in Python (Matplotlib, Seaborn), cutting reporting latency from **1 day to near real-time**, and delivered actionable retention insights that guided targeted campaigns.

Education

University of Southern California, Dornsife GPA: 3.71/4.0
PhD in Computational Neuroscience | Enrolled Fall 2022, expected graduation May 2027
Coursework: CSCI: Deep Learning and its Applications (Pytorch, LLM), DSCI: Machine Learning (Data Mining), EE: Computing Principles (C++, Bazel), ECE: Probability, EE: Data Analysis & Neurotech Design (Pytorch), PSYC: Decision Neuroscience (Bayesian Modeling)

Emory University, Emory College GPA: 3.75/4.0
BS in Quantitative Statistical Science and Computer Science | 2018 - December 2021

Publications

Fang, Xiaotong and Payam Piray (2025). "Inferring the causes of noise from binary outcomes: A normative theory of learning under uncertainty". In: *Psychological Review*. Under review. URL: https://doi.org/10.31219/osf.io/vuc5g_v1.

Zhang, X., S. Ivanovic, B. Moore, X. Fang, and D. Song (2025). "Topological Masking and Attention for Universal Decoding of Spiking Neural Dynamics". In: *NeurIPS '25*. URL: https://openreview.net/forum?id=A6WQTfLqJ7.

Skills

Programming	Python, R, Java, JavaScript, SQL, Scala, C++, MATLAB; Git/GitHub, Spark
ML & AI	Predictive modeling, generative models, sequence models (HMMs, RNNs, GRU, LSTM), transfer learning, RL, Bayesian inference/optimization; PyTorch, TensorFlow
MLOps	Pandas, NumPy, Scikit-learn, ETL pipelines, preprocessing workflows, reproducible artifacts, MLflow, AWS
Analytics	A/B testing, randomized designs, GLMs/mixed-effects, causal inference, hypothesis testing; CV + simulation-based eval
Community	Teaching Assistant (USC, led two sections); Treasurer, Young Researchers Program (managed funding/logistics)